Knowledge Workspace

Semantic PDF Search and Research Workspace

React + FastAPI + Supabase + Docling

Built as a research-focused knowledge system for reading, uploading, indexing, and retrieving PDF content securely.

1. Project Vision

The goal was to build a research workspace that helps users:

- upload and organize PDF documents,
- convert them into searchable content,
- search by meaning, not just exact keywords,
- keep documents and results scoped to a single authenticated user,
- read PDF content with page-aware provenance and context.

Core value

Product intent

2. Original Product Goals

First Milestone goals

- Build a premium reading experience with a calm, research-focused UI.
- Add library, reader, search, notes, login, and settings screens.
- Keep the product clearly marked as a demo.
- Prepare the app for future Supabase integration, embeddings, and AI features.

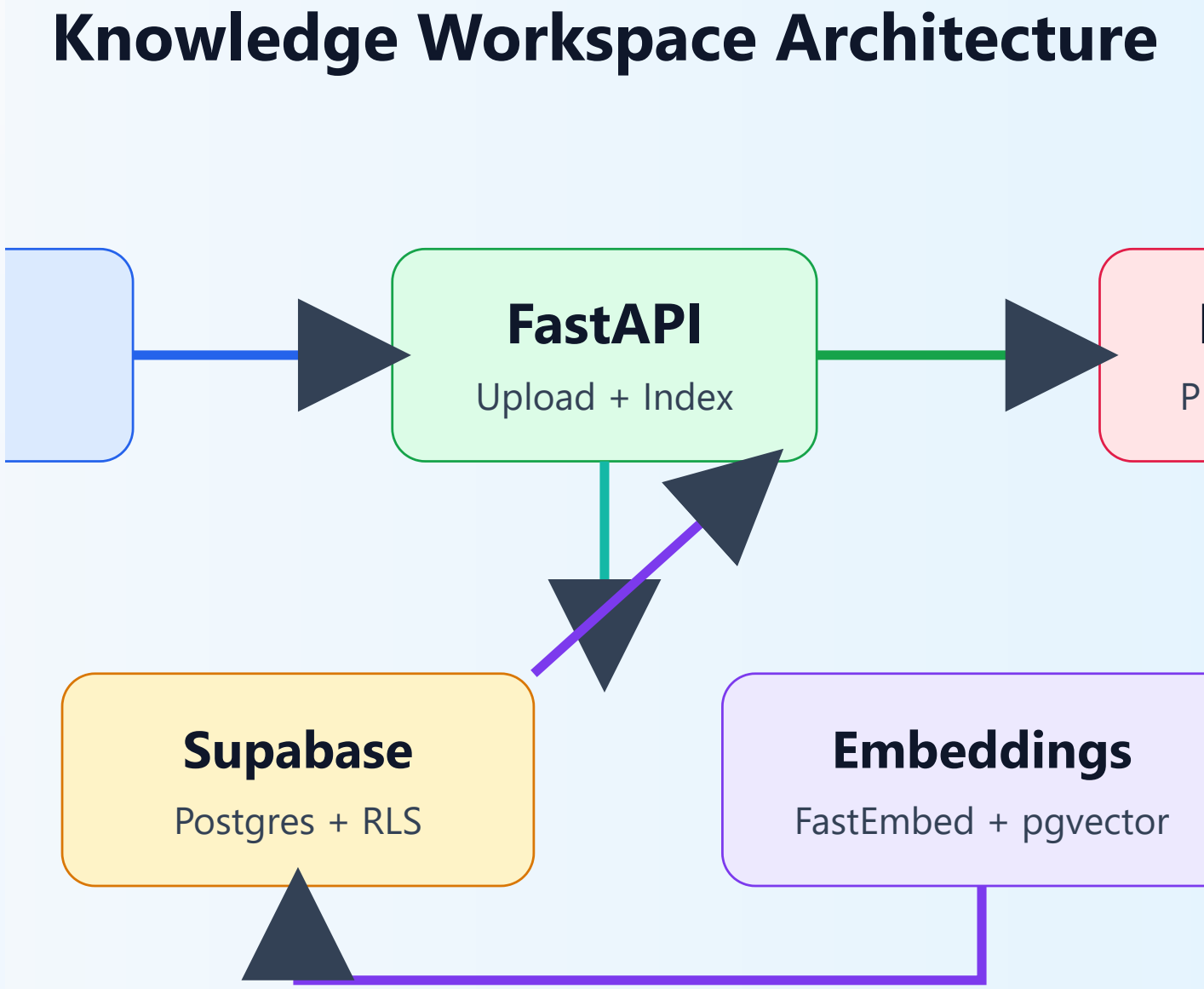
Current ingestion scope

- Upload PDFs from the Library

3. System Architecture

End-to-end flow

1. User uploads a PDF in the frontend.
2. FastAPI validates and stages the file.
3. Docling Serve converts the PDF asynchronously.
4. Extracted content is normalized and chunked.



4. Key Technical Stack

Frontend

- React + TypeScript + Vite
- Search UI and PDF reader shell
- Library, global search, notes, settings

Backend

- FastAPI service for upload, indexing, and search
- Async processing pipeline
- Validation and job state tracking

Data layer

- Supabase Auth
- Postgres + pgvector
- Storage bucket for private PDFs
- RLS for user scope

Supporting tools

5. Retrieval and Search Architecture

- Semantic vector retrieval is performed through pgvector.
- Full-text search uses Postgres `tsvector` and GIN index.
- Exact phrase matching adds a strong retrieval signal.
- Reciprocal Rank Fusion combines all three methods.

Hybrid retrieval model

$$\mathrm{RRF}(d) = \sum_{m \in M} \frac{1}{k + \mathrm{rank}_m(d)}$$

This gives a unified ranking score from:

6. User Experience Flow

From upload to answerable query

1. Open the Library.
2. Drop or select a PDF.
3. Wait for conversion and indexing.
4. Search in the global search dialog.
5. View excerpts with page and provenance metadata.
6. Open the reader and navigate to relevant pages.

Security model

Upload workflow improvement

- `isUploading` was moved to cover only the transfer phase.
- Indexing runs independently per file.
- Multiple uploads can proceed without one blocking the other.

Search deduplication fix

- candidate pool increased before final slicing,
- exact normalized duplicates were removed,
- shared-prefix passages were preserved correctly,
- top results remained non-duplicate and count-correct.

Issues discovered

- unsupported queries still returned results,
- relevance was weak on some broad or unrelated questions,
- there was no reliable labeled evaluation set.

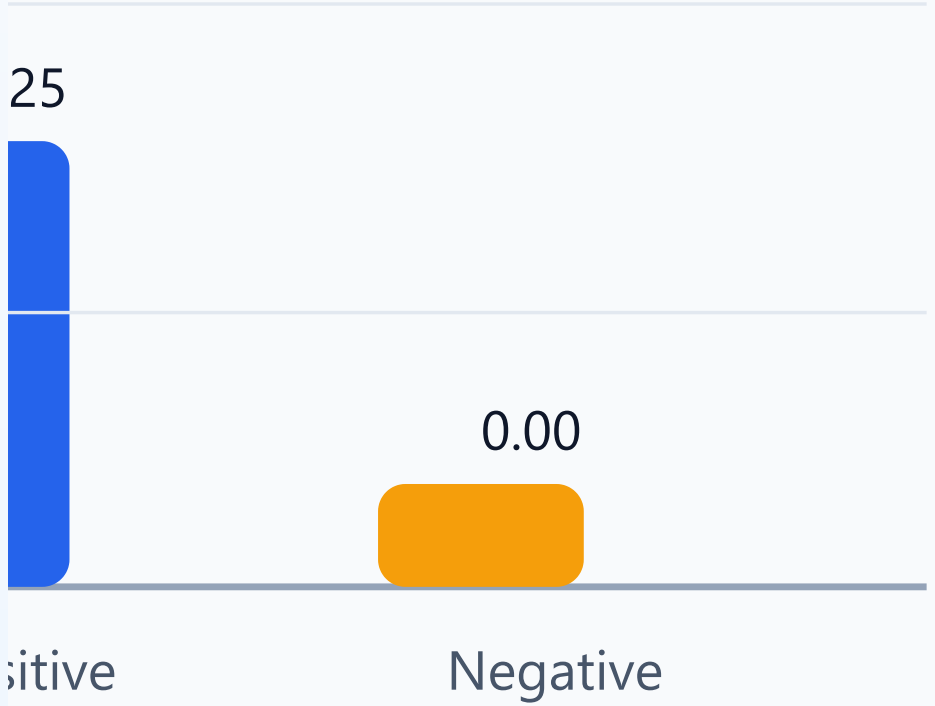
Evaluation approach

- synthetic corpus created for

Search Quality Snapshot

Key f

- Dedu
- Relev
- False



9. Search Quality Metrics

Query Type	#Queries	Precision@5	MRR	False Positive Rate
Positives	8	0.25	0.94	—
Negatives	4	0.00	0.00	100%

Interpretation

- Most correct answers appeared at rank 1.
- But many unsupported queries still produced results.
- This shows the need for a stronger relevance gate, reranker, or abstention policy.

Important takeaway

Local setup checklist

```
docker compose -f backend/docker-compose.yml up -d
cd backend
npm run backend:dev
cd ../src
npm run dev
```

Health checks

- Frontend: `http://localhost:5175`
- Backend: `http://127.0.0.1:8000/api/health`
- Docling: `http://localhost:5001`

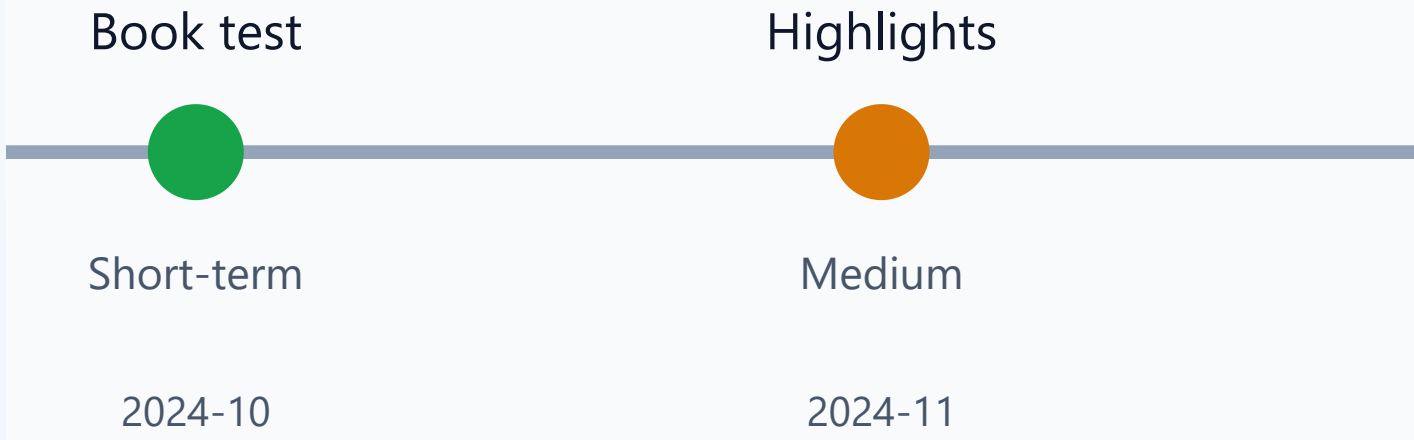
Roadmap

11. Roadmap and Next Steps

Short term

- Real-document evaluation on actual book PDFs
- Search improvements and query validation
- Optional reranker experiment with GPU support

Medium term



12. Final Takeaways

- This project successfully combines a strong UI, secure storage, structured ingestion, and hybrid retrieval.
- The architecture is sound and modular enough for future improvements.
- Search quality is the main remaining technical challenge.
- The next milestone is not just more features, but better relevance and trustworthiness.

Closing message

Knowledge Workspace is already a strong prototype foundation for an AI-powered research assistant, with a clear path from PDF

Thank You

Questions?

- Project architecture
- Search quality and retrieval strategy
- Future enhancements
- Demo walkthrough and next milestones